Crossing Labelling Guide

How to draw boxes in Roboflow from now on. Four classes, two questions per person, and a box around everyone you can see.

The four classes

Type the names exactly as shown: lowercase, with an underscore. The colours match the boxes the app draws.

id 0	id 1	id 2	id 3
`person_left`	`person_right`	`wheeled_left`	`wheeled_right`
On foot, moving towards the left edge of the photo	On foot, moving towards the right edge of the photo	A person with a bike or scooter, moving left	A person with a bike or scooter, moving right

Ask two questions for every person

QUESTION 1

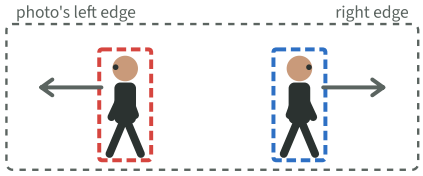
Do they have a bike or scooter?



- **Yes**, riding it, walking beside it, or carrying it: `wheeled_*`. Draw **one box** around the person *and* the vehicle.
- **No**: `person_*`, a box around the person.
- A bike with nobody at it gets **no box**.

QUESTION 2

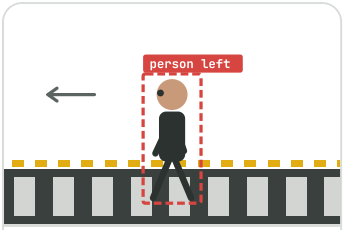
Which way across the photo are they going?



- Judge it **as the camera sees it**, not from the person's own left or right.
- Heading towards the photo's left edge: **_left**. Towards the right edge: **_right**.
- Walking straight towards or away from the camera: pick the side they are **drifting** to. Use their feet and the way their body is angled.
- **Everyone** gets left or right, including people standing still. There is no "other" class.

Examples

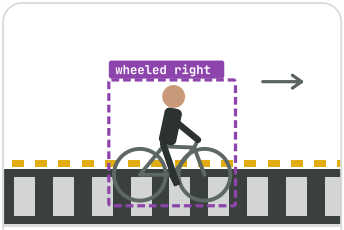
Dashed boxes show what to draw. A red cross marks a mistake to avoid.



DO · WALKING LEFT

`person_left`

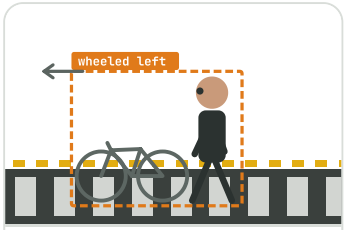
Body and feet point towards the photo's left edge.



DO · RIDING A BIKE

`wheeled_right`

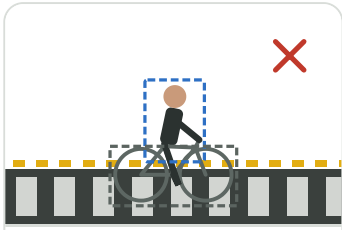
One box: top of the head to the bottom of both wheels.



DO · WALKING A BIKE

`wheeled_left`

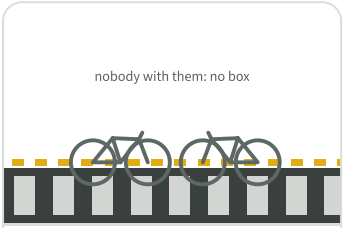
Still wheeled. Box the person and the bike together.



DON'T · PERSON BOX + BIKE BOX

`Separate boxes`

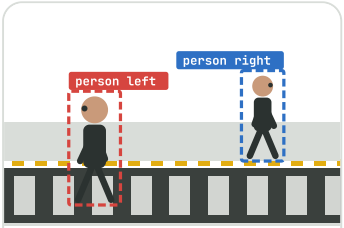
There is no bicycle class. Merge them into one wheeled box.



LEAVE · PARKED BIKE

`no box`

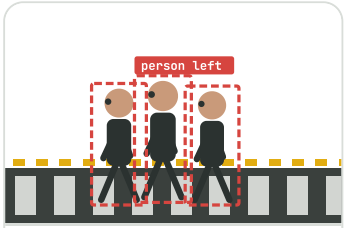
A bike or rack with nobody at it is left unlabelled.



DO · OFF THE CROSSING

`person_right`

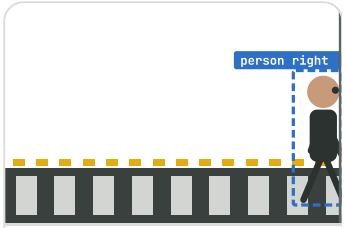
People on the pavement still get a box. The crossing filter is in code.



DO · CROWD

`person_left ×3`

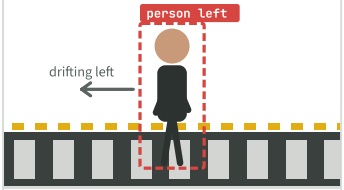
One box per person, even when the boxes overlap.



DO · CUT OFF AT THE EDGE

`person_right`

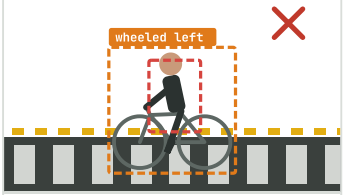
Box the visible part only, right up to the photo edge.



DO · WALKING TOWARDS THE CAMERA

`person_left`

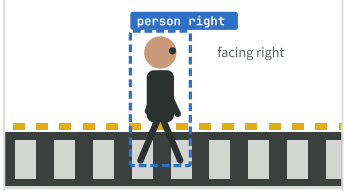
Pick the side they drift to. This one is drifting left.



DON'T · PERSON BOX INSIDE WHEELED

`Box inside a box`

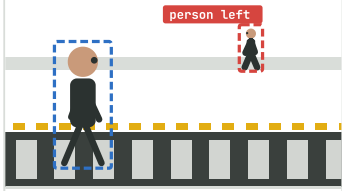
The rider is already inside the wheeled box. Delete the extra one.



DO · STANDING AND WAITING

`person_right`

Still labelled. Use the way they face or are about to cross.



DO · SMALL AND FAR AWAY

`person_left`

Zoom in. If you can tell it is a person, it gets a box.

Quick reference

YOU SEE	LABEL	BOX COVERS
Someone walking towards the left of the photo	`person_left`	Head to feet
Someone walking towards the right of the photo	`person_right`	Head to feet
Someone riding a bike or e-scooter to the left	`wheeled_left`	Rider and whole vehicle, wheels included
Someone walking or carrying a bike to the right	`wheeled_right`	Person and bike together
Someone standing still, waiting	`person_left` / `person_right`	Pick the way they face or are about to go
Someone off the zebra crossing (pavement, far side)	Label as normal	Counting only the crossing is done in code, not by leaving people out
Someone cut off by the photo edge or behind a pole	Label as normal	Only the part you can see
A parked bike, or a bike rack, with nobody at it	`no box`	
A person on a poster, a reflection, a statue	`no box`	

Drawing the box

Do

- **Tight edges.** Touch the top of the head, the soles of the feet, and the furthest arm or bag.
- **One box per person**, even in a crowd where boxes overlap.
- **Small, far-away people count.** Zoom in. If you can tell it is a person, label it.
- **Hidden people:** box only the visible part. Do not guess where hidden legs would be.
- **Wheeled boxes** run from the top of the rider's head to the bottom of the lowest wheel, and from the front wheel to the back wheel.

Never

- Leave a visible person without a box. The model learns "this is not a person" and starts missing people.
- Use a `bicycle`, `cyclist` or `person_other` class. Only the four above exist.
- Draw a person box **inside** a wheeled box, or a separate box for the bike.
- Turn on flip augmentation in Roboflow. A mirrored person going left is labelled left but is really going right. Leave all augmentation off: training adds its own.
- Upload photos anywhere public. Keep them in the private team project only.

In Roboflow

While labelling

1. Work in the private team project, `c389_project_final`.
2. Tag every upload with where and when it was shot, e.g. `crossing_A_2026-11-05`, or `outside_data` for images not taken by us.
3. Check the **layers list** on each image. A tiny stray box is easier to find there than on the picture.
4. Pre-labelled images arrive with guessed directions. Check every person's direction, not only the flagged ones.
5. The train / valid / test split in Roboflow does not matter. We re-split every export 70/20/10 in code so near-identical photos never land in different splits.

After each export (YOLO26 format)

Send the zip to whoever runs the scripts. They run, from the repo:

```
python -m src.data.remap_classes
EXPORT --out FIXED
python -m src.data.resplit FIXED --
out SPLIT --ratio 70/20/10
python -m src.data.verify_export
SPLIT --expect 4
python -m src.data.dedupe_check
SPLIT
```

An export that still has a `bicycle` class goes through `src.data.to_wheeled` instead of `remap_classes`.

Team rule, agreed 3 Nov 2026: walking or carrying a bike counts as wheeled. Classes follow the lecturer's Twist 2 table. Questions about an odd case: post a screenshot in the group chat before guessing, and add the decision to this guide.